

# Divide Fractions

Lesson 2-3

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*In  $3/4 \div 1/2$ , the dividend is  $3/4$  — the total amount being split*

### Dividend

The number you are splitting up in a division problem.

*In  $3/4 \div 1/2$ , the divisor is  $1/2$  — the size of each group*

### Divisor

The number you split by in a division problem.

*The reciprocal of  $2/3$  is  $3/2$ . To divide by  $2/3$ , multiply by  $3/2$ .*

### Reciprocal

A fraction turned upside down.

*In  $3/4 \div 1/4 = 3$ , the quotient is 3 — three  $1/4$ -size pieces fit into  $3/4$*

### Quotient

The answer when you divide.

*$8/12 \rightarrow \text{GCF is } 4 \rightarrow 8 \div 4 / 12 \div 4 = 2/3$*

### Simplify

To make a fraction smaller using the same parts, like  $2/4 = 1/2$ .

## Key Ideas & Notes



### WHAT WE'RE LEARNING TODAY

**I can divide a fraction by a fraction by multiplying by the reciprocal and simplifying.**

#### 1 Notice

Agent Rivera is dividing a  $\frac{3}{4}$ -pound evidence bag into portions that each weigh  $\frac{1}{8}$  pound.

#### 2 Key idea

I can divide a fraction by a fraction by multiplying by the reciprocal and simplifying.

#### 3 Words I'll use

Dividend

Divisor

Reciprocal

Quotient

Simplify



#### Think About It

- What is the total weight being divided?
- What is the size of each portion?
- Will the answer be more or less than 1?



#### Watch

Watch your teacher model one example. Jot what you see.

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#### We try

Solve the next one together as a class.

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#### You try

Now try one on your own.

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Agent Rivera splits a  $\frac{3}{4}$ -pound evidence bag into  $\frac{1}{8}$ -pound portions. Will she get more or fewer than 1 portion? Estimate before solving.**

#### Level 1 support

**Start here:** She will get many portions because  $\frac{1}{8}$  pound is much smaller than  $\frac{3}{4}$  pound.

 **Need a hint?**

**Sentence starters (optional)**

**She will get \_\_\_\_ than 1 portion because \_\_\_\_.**

*Ella obtendrá \_\_\_\_ que 1 porción porque \_\_\_\_.*

**Since  $\frac{3}{4}$  is the dividend and  $\frac{1}{8}$  is the divisor, \_\_\_\_.**

*Como  $\frac{3}{4}$  es el dividendo y  $\frac{1}{8}$  es el divisor, \_\_\_\_.*

**Word bank:** dividend divisor quotient

**Listen for:** Students recognize the divisor  $\frac{1}{8}$  is smaller than the dividend  $\frac{3}{4}$ , so the quotient is greater than 1.

#### Level 2

**Estimate the exact number of portions and explain your reasoning using equivalent fractions before computing.**

- I estimate about \_\_\_\_ portions because  $\frac{3}{4}$  equals \_\_\_\_ eighths.
- Comparing eighths to eighths shows \_\_\_\_.

## EXPLORE

### Why do we multiply by the reciprocal when we divide one fraction by another?

#### Level 1 support

**Start here:** Multiplying by the reciprocal undoes the divisor and gives the same result as dividing.



**Need a hint?**

**Sentence starters (optional)**

**I flipped the divisor to its reciprocal because \_\_\_\_.**

*Volteé el divisor a su recíproco porque \_\_\_\_.*

**After Keep, Change, Flip the answer is \_\_\_\_.**

*Después de Mantener, Cambiar, Voltear la respuesta es \_\_\_\_.*

**Word bank:** reciprocal divisor quotient simplify

**Listen for:** Students explain that flipping the divisor turns division into multiplication, and they simplify the final quotient.

#### Level 2

**Show why  $3/4 \div 1/8$  gives the same answer whether you count eighths in  $3/4$  or multiply by the reciprocal.**

- Both methods agree because \_\_\_\_.
- Counting eighths matches multiplying by 8 because \_\_\_\_.

CONNECT

How could dividing fractions help you share food or materials fairly?

Level 1 support

**Start here:** Dividing fractions tells you how many equal fraction-size shares fit into an amount.



Need a hint?

**Sentence starters (optional)**

**Dividing fractions helps when \_\_\_\_.**

*Dividir fracciones ayuda cuando \_\_\_\_.*

**For example, \_\_\_\_ of a \_\_\_\_ split into \_\_\_\_ shares gives \_\_\_\_.**

*Por ejemplo, \_\_\_\_ de un \_\_\_\_ dividido en \_\_\_\_ partes da \_\_\_\_.*

**Word bank:** **dividend** **divisor** **simplify**

**Listen for:** Students give a fair-share scenario and identify which fraction is the dividend and which is the divisor.

Level 2

**A classmate flipped the dividend instead of the divisor. Critique this error and explain the correct step.**

- Flipping the dividend is wrong because \_\_\_\_.
- Only the \_\_\_\_ should be flipped because \_\_\_\_.

## Guided Examples

### Example 1

**What is the reciprocal of  $\frac{3}{5}$ ?**

**Solution:** To find the reciprocal, flip the fraction. The reciprocal of  $\frac{3}{5}$  is  $\frac{5}{3}$ .

**Answer:** A.  $\frac{5}{3}$

### Example 2

**What is  $\frac{1}{2} \div \frac{1}{4}$ ?**

**Solution:**  $\frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1} = \frac{4}{2} = 2$ . Two  $\frac{1}{4}$ -size pieces fit into  $\frac{1}{2}$ .

**Answer:** A. 2

### Example 3

**What is  $\frac{3}{4} \div \frac{1}{8}$ ?**

**Solution:**  $\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times \frac{8}{1} = \frac{24}{4} = 6$ . Six  $\frac{1}{8}$ -size pieces fit into  $\frac{3}{4}$ .

**Answer:** A. 6

## Write About the Math

### The Writing Revolution

I can explain my steps using the words dividend, divisor, reciprocal, quotient, and simplify.

#### 1. Kernel Sentence subject + verb

**Model:** Simplify is to make a fraction smaller using the same parts, like  $2/4 = 1/2$ .  
*Simplificar es hacer una fracción más pequeña con las mismas partes, como  $2/4 = 1/2$ .*

**Write a kernel sentence about simplify. Use a subject and a verb.**

*Escribe una oración base sobre simplificar. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because · but · so

**Kernel:** Simplify matters in math  
*Simplificar importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Simplify matters in math because \_\_\_\_.**  
*Simplificar importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Simplify matters in math, but \_\_\_\_.**  
*Simplificar importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Simplify matters in math, so \_\_\_\_.**  
*Simplificar importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

**Statement**  
*Afirmación*

Tell one true fact about simplify.  
*Di un hecho verdadero sobre simplify.*

**Simplify \_\_\_\_.**

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**Question**  
*Pregunta*

Ask a question about simplify.  
*Haz una pregunta sobre simplify.*

**How does \_\_\_\_ ?**

*¿Cómo \_\_\_\_ ?*

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**Exclamation**  
*Exclamación*

Show excitement about simplify.  
*Muestra entusiasmo sobre simplify.*

**Wow, \_\_\_\_ !**

*¡Guau, \_\_\_\_ !*

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**Command**  
*Mandato*

Tell a partner what to do with simplify.  
*Dile a un compañero qué hacer con simplify.*

**First, \_\_\_\_ .**

*Primero, \_\_\_\_ .*

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### 4. Explain Your Reasoning use a sentence starter

**She will get \_\_\_\_ than 1 portion because \_\_\_\_.**

*Ella obtendrá \_\_\_\_ que 1 porción porque \_\_\_\_.*

**Since  $\frac{3}{4}$  is the dividend and  $\frac{1}{8}$  is the divisor, \_\_\_\_.**

*Como  $\frac{3}{4}$  es el dividendo y  $\frac{1}{8}$  es el divisor, \_\_\_\_.*

**I flipped the divisor to its reciprocal because \_\_\_\_.**

*Volteé el divisor a su recíproco porque \_\_\_\_.*

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## Try It

Solve on your own. Check the answer key when you are done.

**1. What is  $\frac{2}{3} \div \frac{1}{3}$ ?**

- A. 2
- B.  $\frac{2}{9}$
- C.  $\frac{1}{2}$
- D. 6

Show your work:

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**2. A recipe needs  $\frac{2}{3}$  cup of sugar. You only have a  $\frac{1}{6}$ -cup scoop. How many scoops do you need?**

- A. 4 scoops
- B. 3 scoops
- C. 6 scoops
- D. 2 scoops

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**Is  $\frac{3}{4} \div \frac{1}{2}$  the same as  $\frac{1}{2} \div \frac{3}{4}$ ? Solve both and explain why or why not. What does this tell you about the order in division?**

*Sentence starter:  $\frac{3}{4} \div \frac{1}{2} = \underline{\hspace{1cm}}$  and  $\frac{1}{2} \div \frac{3}{4} = \underline{\hspace{1cm}}$ . These are  $\underline{\hspace{1cm}}$  because  $\underline{\hspace{1cm}}$ . This shows that division is  $\underline{\hspace{1cm}}$ .*

Show your work:

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## Reflect — Exit Ticket

**What is  $\frac{4}{5} \div \frac{2}{5}$ ?**

- A. 2
- B.  $\frac{8}{25}$
- C.  $\frac{2}{5}$
- D.  $\frac{10}{5}$

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** A.  $2 - \frac{2}{3} \div \frac{1}{3} = \frac{2}{3} \times \frac{3}{1} = \frac{6}{3} = 2$ . *Two  $\frac{1}{3}$ -size pieces fit into  $\frac{2}{3}$ .*
2. **Try It 2:** A. 4 scoops —  $\frac{2}{3} \div \frac{1}{6} = \frac{2}{3} \times \frac{6}{1} = \frac{12}{3} = 4$  scoops.
3. **Exit Ticket:** A.  $2 - \frac{4}{5} \div \frac{2}{5} = \frac{4}{5} \times \frac{5}{2} = \frac{20}{10} = 2$ . *Two groups of  $\frac{2}{5}$  fit into  $\frac{4}{5}$ .*

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Simplify is to make a fraction smaller using the same parts, like  $\frac{2}{4} = \frac{1}{2}$ .
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).